

A Biomimetic Tunnel FET-Based Spiking Neuron for Energy-Efficient Neuromorphic Computing With Reduced Hardware Cost

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Abstract—In this work, utilizing the unique features of conventional Si-based tunnel FET (TFET), a TFET-based leaky integrate-and-fire (LIF) neuron with higher energy efficiency and reduced hardware cost is proposed. Compared with traditional CMOS-based LIF neuron, the proposed TFET-based LIF neuron can produce an additional bio-plausible after-hyperpolarization (AHP) behavior and relative refractory period without extra hardware cost by exploiting the features of large Miller effect and forward p-i-n current in TFET. Moreover, the typical ambipolar effect and superlinear onset behaviors in conventional Si-based TFET enable the lower hardware cost and lower energy consumption ($\sim 10\times$ reduction) for TFET-based neuron. Furthermore, the proposed TFET neuron-based spiking neural network (SNN) is demonstrated for pattern recognition tasks, showing its advantage of significant energy efficiency. This work provides a promising highly integrated and energy-efficient solution for the hardware implementation of spiking neuron for neuromorphic computing.

Index Terms—Biomimetic spiking neuron, neuromorphic computing, relative refractory period, tunnel FET (TFET).

I. INTRODUCTION

BIO-MIMETIC spiking neural networks (SNNs) [1] show great potential of high energy efficiency for intelligence tasks. Traditional CMOS have been widely used for the hardware implementation of leaky integrate-and-fire (LIF) spiking neuron [2]. A RC parallel circuit formed by a fixed-bias nMOS and a capacitor is usually used for leaky integration.

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A CMOS-based amplifier and reset circuit is used to realize spike generation, where the amplifier is usually formed by two serial inverters [2]. However, these neuron circuits show simple functionality behavior, and much more hardware overhead are required when implementing advanced biomimetic functions like after-hyperpolarization (AHP) behavior and refractory period which have been proved to be of great significance to the firing frequency adaption of biological neuron [2]–[5]. Besides, the energy consumption dominated by short-circuit current is also large [2]. For SNN, various emerging memory devices, such as RRAM, have been applied to develop the artificial synapses for significantly improved energy and area efficiency [3], [4], while the artificial neurons are still mostly CMOS-implemented and still suffer from high energy consumption and high hardware cost, making it a critical issue for the large-scale integrated neuromorphic computing systems. One possible solution is to take the advantage of novel beyond-CMOS devices [6], [7]. Tunnel FET (TFET) is a novel steep subthreshold swing device, which is considered as one of the most promising energy-efficient devices [8], [9] and can be used for low-power neuromorphic computing applications [10]. Recently, heterojunction TFET based on direct bandgap III-V materials is utilized and simulated for hardware implementation of spiking neuron [10], [11]. Although the supply voltage for neuron can be reduced due to the steep subthreshold swing of TFET, they suffer from high-static-energy-consumption issue due to the high OFF-state leakage current of III-V heterojunction TFET, which can be even more serious for event-driven brain-inspired neuromorphic systems. Meanwhile, the hardware cost for the implementation of III-V-TFET-based neuron is still high since the circuit topology is not changed compared with conventional MOSFET-based neuron [10].

In this work, a spiking neuron based on conventional Si TFET is proposed, achieving higher energy efficiency and lower hardware cost than CMOS implementation, as well as an additional bio-plausible AHP behavior. The SNN based on our proposed neuron is also implemented, realizing pattern recognition with high energy efficiency. This brief is organized as follows. Section II presents the conventional Si TFET device characteristics for spiking neuron. Section III explains the TFET-based neuron design, and the functionality and neuronal

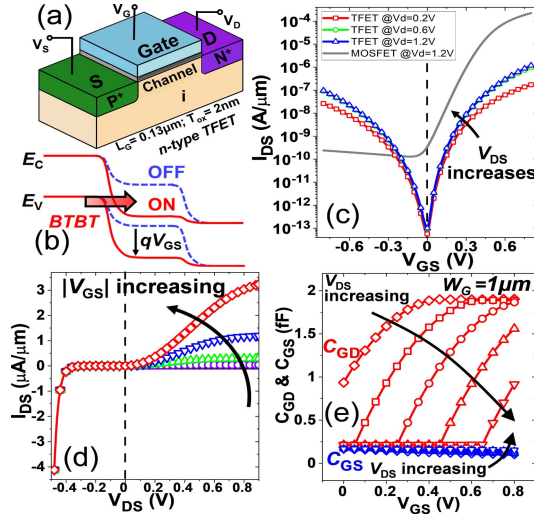


Fig. 1. (a) Device structure of Si n-type bulk tunnel FET (TFET). The doping concentrations of the p-type source, p-type channel, and n-type drain are 1×10^{20} , 5×10^{14} , and $1 \times 10^{20} \text{ cm}^{-3}$, respectively in this work. (b) Schematic of the OFF-state and ON-state surface energy bands along the channel direction of the Si n-type TFET. (c) Transfer curves show the ambipolar effect and fundamental ultralow OFF-state reverse p-i-n leakage current. The MOSFET behavior is based on the 0.13- μm technology node. (d) Output curves show the superlinear on-set and forward p-i-n current in negative V_{DS} . (e) C_{GS} and C_{GD} is significantly larger than C_{GS} .

energy consumption is analyzed. Section IV demonstrates the SNN based on proposed TFET neuron for pattern recognition. Finally, the conclusion is presented in Section V.

II. TFET DEVICE USED FOR NEURON

Fig. 1 shows the simulation results of conventional Si TFET devices based on our previously developed surface-potential-based analytical current model [12] and capacitance model [13], which includes detailed physical analysis and corresponding formula description. Based on [12] and [13], the Verilog-A models for Si TFET are established for the SPICE circuit stimulation in this work. The device models have been calibrated with experimental results [9] and can accurately reflect the typical features of conventional TFET. For a typical TFET device with gated p-i-n structure, it shows the ambipolar band-to-band tunneling current [see Fig. 1(c)], and superlinear onset behavior in output curves with forward p-i-n current behavior in negative V_{DS} region [see Fig. 1(d)] [12]. Compared with MOSFET, TFET shows large Miller capacitance [see Fig. 1(e)] [13]. Based on our TFET models, SPICE simulation of neuron circuit is carried out in this work.

III. TFET-BASED SPIKING NEURON

In this part, the new circuit topology design of TFET-based LIF neuron is proposed, and the functionality and energy consumption of the neuron are analyzed by SPICE simulation.

A. Circuit Topology Design of TFET-Based Neuron

Based on the Si TFET device mentioned above, a novel TFET-based LIF neuron circuit is proposed. As shown

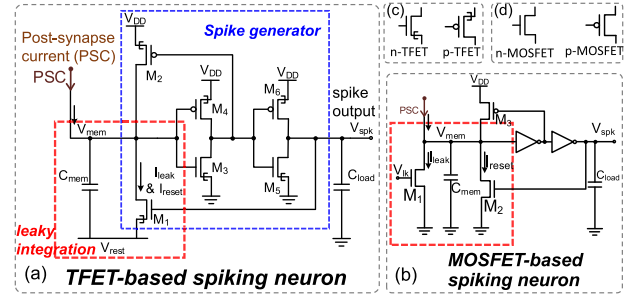


Fig. 2. (a) Schematic of the proposed TFET-based spiking neuron, consisting of two parts: a leaky integrate part and a spike generator. (b) Schematic of MOSFET-based neuron. The graphic symbol of (c) TFET and (d) MOSFET.

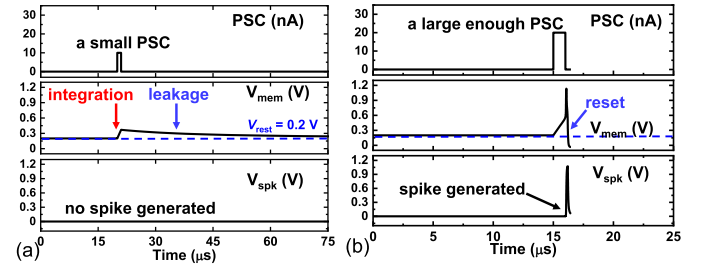


Fig. 3. Behavior of proposed TFET neuron based on SPICE simulation. (a) Subthreshold behavior with a small input PSC applied. (b) Spike firing process with strong PSC applied. V_{mem} is the membrane potential, V_{spk} is the output voltage, and V_{rest} is the resting potential.

in Fig. 2, the proposed neuron circuit consists of two parts as the LIF model describes. One is a leaky integrate part, which consists of a capacitor C_{mem} modeling the membrane of biological neuron and a transistor M_1 for leakage path, realizing the function of integrating the input post-synapse current (PSC). The other one is a spike generator, which consists of two serial inverters (M_3 – M_6) for threshold comparison, one transistor (M_1) for reset path, and another one (M_2) for positive feedback, realizing the function of generating the output spike. Due to the ambipolar effect of TFET, the leakage and reset path of the TFET neuron can be simultaneously realized by one single transistor of M_1 . As shown in Fig. 2(a) and (b), compared with traditional MOSFET-based LIF neuron, a separate transistor for leakage path can be eliminated due to functional reuse of M_1 in TFET neuron, and thus reduce the hardware cost.

SPICE simulation is carried out to verify the basic LIF behavior. Fig. 3 shows the leaky integration process of the TFET-based neuron. Without any PSC, the membrane potential V_{mem} will be equal to the resting potential V_{rest} , which is slightly higher than the ground voltage (e.g., $V_{DD} = 1.2 \text{ V}$, $V_{rest} = 0.2 \text{ V}$). Therefore, when there is no spike generated, the V_{gs} bias of M_1 is negative. With PSC applied, the membrane potential V_{mem} will increase due to the charge integration in the capacitor (C_{mem}). Without subsequent PSC applied, V_{mem} will drop to the resting potential V_{rest} due to leaky effect caused by ambipolar current of M_1 [see Fig. 3(a)]. At this time, M_1 provides a leakage path.

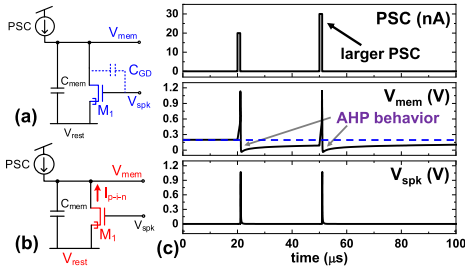


Fig. 4. AHP behavior of the TFET neuron ($C_{mem} = 50$ fF; gate-width of M_1 is $10\ \mu\text{m}$, which depends on the required capacitor value and C_{GD} per unit area of TFET, and is comparable with M_2 in MOSFET-based neuron). The schematic and principle of (a) AHP process due to the large C_{GD} of TFET, and (b) AHP recovery process due to the forward p-i-n current of M_1 . (c) Simulated AHP behavior of proposed TFET neuron.

With strong PSC applied [see Fig. 3(b)], the V_{mem} will reach the threshold of the spike generator, and the output voltage V_{spk} will rapidly increase. M_2 provides extra positive feedback current to make the V_{mem} continuously increase over the threshold. At this time, the gate bias of M_1 becomes positive and the capacitor will be rapidly discharged through M_1 . Therefore, M_1 provides a reset path. Then the V_{spk} will return to ground voltage, forming a complete spike. Therefore, during the LIF process, M_1 can serve as both leakage and reset path due to the ambipolar effect, reducing the hardware cost of LIF neuron.

B. Advanced Functionality of TFET-Based Neuron

Besides the basic LIF function, the proposed TFET-based neuron can further achieve advanced biomimetic behavior. In biological neuron, the membrane potential usually overshoots the resting potential at the end of an action potential duration, and then gradually recovers. This behavior is called AHP, which causes relative refractory period and has been proven to be of great significance to the firing frequency adaption of biological neuron [5]. However, it requires extra capacitor-based circuits ($6T + 1C$) for this function for CMOS-neuron [2].

Our proposed TFET-based LIF neuron can additionally produce the bio-plausible AHP behavior without extra hardware. As shown in Fig. 4, at the end of a spike firing, due to the large gate to drain capacitance C_{GD} of TFET, the sharp dropping of the output voltage can be coupled to the V_{mem} by large Miller capacitance, which makes the V_{mem} further drop below the V_{rest} , generating the AHP behavior. The drain bias V_{ds} of M_1 becomes negative, and the V_{mem} will recover to the resting potential due to the charging process of C_{mem} caused by the forward p-i-n current of M_1 , imitating the AHP recovery process and thus the relative refractory period in biological neurons. During this period, a stronger stimulus is needed to trigger an action potential, similar to biological neurons.

C. Energy Consumption of TFET-Based Neuron

The energy consumption per spike of the TFET neuron is analyzed and compared. Fig. 5(a) and (b) shows the simulated

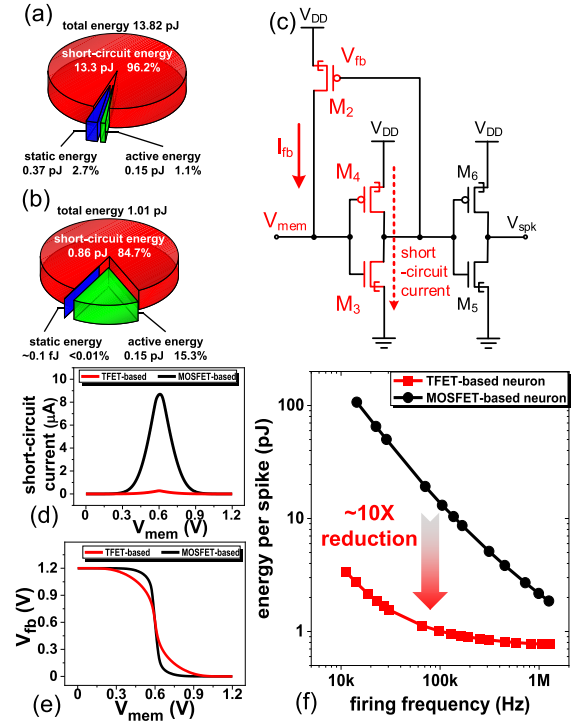


Fig. 5. Composition of energy consumption during per spike firing process at firing frequency of 100 kHz of (a) MOSFET-based neuron and (b) TFET-based neuron. (c) Schematic of the dominant short-circuit current flow path and the positive feedback loop of M_2 in the spike generator of the TFET neuron. (d) Short-circuit current and (e) voltage transfer curves of the first stage inverter (M_3 – M_4) based on TFET and MOSFET. (f) Comparison of energy per spike between TFET-based and MOSFET-based neuron with different firing frequency.

energy consumption of the proposed TFET-based neuron and conventional MOSFET-based neuron as shown in Fig. 2. The short-circuit energy dominates the total energy consumption. Compared with MOSFET-based neuron with the same active energy, the TFET neuron shows significantly reduced static energy and short-circuit energy and thus the total energy consumption. The static energy reduction can be ascribed to the ultralow OFF current of TFET. As shown in Fig. 5(c), the short-circuit energy is dominated by the first stage inverter in the spike generator, which is proportional to the short-circuit current and short-circuit time. Fig. 5(d) shows the dependence of short-circuit current of the first stage inverter on the membrane potential V_{mem} in both TFET-based neuron and MOSFET-based neuron. The short-circuit current in TFET-based neuron is much smaller due to smaller drive current of TFET than that of MOSFET. On the other hand, the superlinear onset behavior of TFET makes the voltage-transfer characteristics of TFET inverter more gradual than that of MOSFET inverter [see Fig. 5(e)], which can enhance the positive feedback of M_2 during neuronal integrate process by turning on M_2 earlier and accelerating the integrate process of V_{mem} . Therefore, the short-circuit time of the TFET-based neuron can be effectively reduced. For these two reasons, the TFET-based neuron shows smaller short-circuit energy consumption than MOSFET-based neuron.

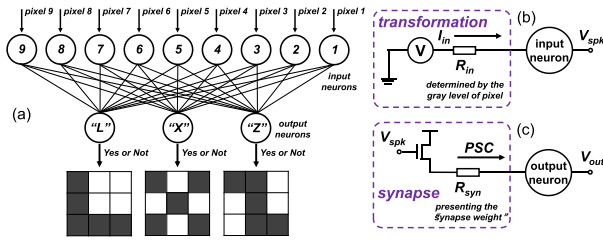


Fig. 6. (a) Structure of spiking neural network (SNN) based on proposed TFET-based neuron for 3×3 -pixel pattern recognition. (b) Transformation of the pixel's gray level to neuronal input current, and (c) the synapse in SPICE circuit simulation of the SNN. The pixel's gray level (c).

In addition, the firing frequency of neuron can further affect the energy consumption per spike due to its impact on the short-circuit time. Fig. 5(f) shows the comparison of energy per spike with different firing frequency. As a result, the TFET-based neuron can achieve significant energy saving compared with the MOSFET-based neuron with different firing frequency.

IV. SNN BASED ON PROPOSED TFET NEURON FOR PATTERN RECOGNITION

A simplified 3×3 -pixel pattern recognition SNN system consisting of nine input spiking neurons and three output neurons is constructed based on our proposed TFET-based neuron, which mainly focus on demonstrating the operating process of SNN including pixel-signal transformation, synapse weight connections and proposed TFET neurons, and evaluating the energy efficiency at SNN system level. As shown in Fig. 6(a), the neural network is designed to recognize three pixel-patterns of capital letters "L," "X," and "Z." The synapse weights in the demonstrated neural network are offline trained and the network is responsible for online inference. The inference function and energy consumption of the SNN is simulated by SPICE. As shown in Fig. 6(b), a voltage source and a resistor are used to code the pixel's gray level, and specifically a gray (/white) pixel corresponds to a voltage source with V_{DD} (/ground voltage). In this work, the synapse between input and output neurons is realized by a 1T1R structure [see Fig. 6(c)]. The output (V_{spk}) of the input neuron is connected to the transistor gate of the synapse and the modulated resistor with low or high resistance is used to control the binary synapse weight, which can determine the synaptic drive current for post neuron. The weight value can be determined by offline-training with spike timing-dependent plasticity (STDP) algorithm [14], and the modulated resistor can also be replaced by some nonvolatile memories such as RRAM [3], [4].

Fig. 7(a) shows the output signals of the output layer neurons with input pixel pattern "Z." The "Z"-output neuron continuously receives strong PSC spikes, and its membrane potential increases rapidly to trigger spike firing. At the same time, the other two output neurons receive weak PSC spikes due to the different synapse weights, resulting in a slower membrane potential increase and thus a later firing, as shown in Fig. 7(a). Therefore, in this case, the different firing order

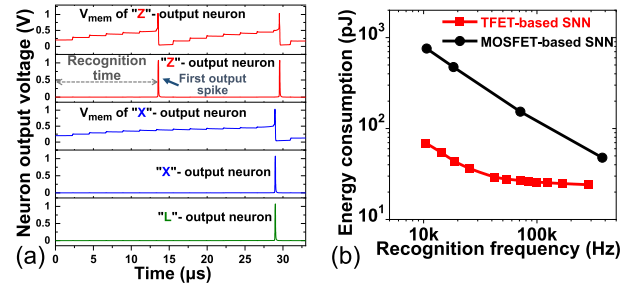


Fig. 7. (a) Output voltage signals of the output-layer neurons with input pixel pattern "Z" in the SNN based on proposed TFET-based neurons. Starting from the pattern input, the "Z"-output neuron is the first to generate output spike among the output layer neurons, indicating the correct recognition. (b) Simulated total energy consumption of the SNN with TFET-based neurons and MOSFET-based neurons. The recognition frequency is the reciprocal of the time from the pattern input to the first spike firing of the output layer neuron, which indicates the operating speed of the SNN.

of the three output neurons indicates that the input pixel pattern "Z" is recognized.

Fig. 7(b) shows the HSPICE simulated total energy consumption of the SNN circuit with our proposed TFET-based neurons and conventional MOSFET-based neurons respectively, which includes energy consumption of all synapses and neurons and is evaluated by multiplying the total inference time by the average power consumption during the pattern recognition inference process. Compared with SNN based on CMOS neuron, the SNN based on TFET neuron can significantly save the energy consumption, which is mainly ascribed to the short-circuit energy reduction as discussed above. Compared with the state-of-art spiking neuron based on MOSFET [2] and TFET [10], [11], our proposed TFET neuron shows the higher energy efficiency and the lower hardware cost, as well as AHP behavior. For more complicated SNNs, it can be anticipated that the static energy consumption may take a larger portion of the total energy cost due to the event-driven feature. Therefore, TFET may further help reduce static energy consumption in large-scale SNNs due to its ultralow OFF current. In conclusion, TFET can bring great advantages of energy efficiency to the hardware implementation of SNNs and can be considered a promising device candidate for the novel non-Von Neumann neuromorphic computing system.

V. CONCLUSION

In this work, a TFET-based spiking neuron is proposed. Compared with traditional MOSFET-based neuron, it achieves the lower hardware cost and the higher energy efficiency with reduced dominant short-circuit energy consumption, as well as an additional bio-plausible AHP behavior and relative refractory period. The demonstrated SNN constructed with TFET-based neuron shows significant energy saving. In summary, by utilizing the disadvantages of conventional Si TFET, our proposed TFET neuron properly turns them into advantages and provides a promising solution for the hardware implementation of spiking neuron for energy-efficient and ultralow-hardware-cost neuromorphic computing systems.

REFERENCES

- [1] W. Maass, "Networks of spiking neurons: The third generation of neural network models," *Neural Netw.*, vol. 10, no. 9, pp. 1659–1671, Dec. 1997, doi: [10.1016/S0893-6080\(97\)00011-7](https://doi.org/10.1016/S0893-6080(97)00011-7).
- [2] G. Indiveri *et al.*, "Neuromorphic silicon neuron circuits," *Frontiers Neurosci.*, vol. 5, p. 73, May 2011, doi: [10.3389/fnins.2011.00073](https://doi.org/10.3389/fnins.2011.00073).
- [3] X. Zhang *et al.*, "Fully memristive SNNs with temporal coding for fast and low-power edge computing," in *IEDM Tech. Dig.*, Dec. 2020, pp. 29.6.1–29.6.4, doi: [10.1109/IEDM13553.2020.9371937](https://doi.org/10.1109/IEDM13553.2020.9371937).
- [4] W. Wan *et al.*, "33.1 A 74 TMACS/W CMOS-RRAM neurosynaptic core with dynamically reconfigurable dataflow and *in-situ* transposable weights for probabilistic graphical models," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 498–500, doi: [10.1109/ISSCC19947.2020.9062979](https://doi.org/10.1109/ISSCC19947.2020.9062979).
- [5] J. G. Nicholls, A. R. Martin, P. A. Fuchs, D. A. Brown, M. E. Diamond, and D. A. Weisblat, "Tonic basis of the action potential," in *From Neuron to Brain*, 5th ed. Sunderland, MA, USA: Sinauer Associates, 2012, ch. 7, pp. 114–128.
- [6] C. Chen *et al.*, "Bio-inspired neurons based on novel leaky-FeFET with ultra-low hardware cost and advanced functionality for all-ferroelectric neural network," in *Proc. Symp. VLSI Technol.*, Jun. 2019, pp. T136–T137, doi: [10.23919/VLSIT.2019.8776495](https://doi.org/10.23919/VLSIT.2019.8776495).
- [7] J. Luo *et al.*, "Capacitor-less stochastic leaky-FeFET neuron of both excitatory and inhibitory connections for SNN with reduced hardware cost," in *IEDM Tech. Dig.*, Dec. 2019, pp. 6.4.1–6.4.4, doi: [10.1109/IEDM19573.2019.8993535](https://doi.org/10.1109/IEDM19573.2019.8993535).
- [8] Y. Zhao, Q. Huang, and R. Huang, "A novel tunnel FET design through hybrid modulation with optimized subthreshold characteristics and high drive capability," *Sci. China Inf. Sci.*, vol. 63, no. 2, Feb. 2020, Art. no. 129402, doi: [10.1007/s11432-019-9874-9](https://doi.org/10.1007/s11432-019-9874-9).
- [9] Q. Huang *et al.*, "Self-depleted T-gate Schottky barrier tunneling FET with low average subthreshold slope and high ION/OFF by gate configuration and barrier modulation," in *IEDM Tech. Dig.*, Dec. 2011, pp. 16.2.1–16.2.4, doi: [10.1109/IEDM.2011.6131564](https://doi.org/10.1109/IEDM.2011.6131564).
- [10] D. Rajasekharan, T. Dutta, A. R. Trivedi, and Y. S. Chauhan, "Energy-efficient spiking neural networks based on tunnel FET," in *Proc. 3rd Int. Conf. Emerg. Electron. (ICEE)*, Dec. 2016, pp. 1–4, doi: [10.1109/ICEmElec.2016.8074589](https://doi.org/10.1109/ICEmElec.2016.8074589).
- [11] D. Rajasekharan, S. S. Chauhan, A. R. Trivedi, and Y. S. Chauhan, "Energy and area efficient tunnel FET-based spiking neural networks," in *Proc. IEEE 2nd Electron Devices Technol. Manuf. Conf. (EDTM)*, Mar. 2018, pp. 59–61, doi: [10.1109/EDTM.2018.8421527](https://doi.org/10.1109/EDTM.2018.8421527).
- [12] C. Wu, R. Huang, Q. Huang, C. Wang, J. Wang, and Y. Wang, "An analytical surface potential model accounting for the dual-modulation effects in tunnel FETs," *IEEE Trans. Electron Devices*, vol. 61, no. 8, pp. 2690–2696, Jun. 2014, doi: [10.1109/TED.2014.2329372](https://doi.org/10.1109/TED.2014.2329372).
- [13] J. Wang, C. Wu, Q. Huang, C. Wang, and R. Huang, "A closed-form capacitance model for tunnel FETs with explicit surface potential solutions," *J. Appl. Phys.*, vol. 116, no. 9, Sep. 2014, Art. no. 094501, doi: [10.1063/1.4894624](https://doi.org/10.1063/1.4894624).
- [14] P. U. Diehl and M. Cook, "Unsupervised learning of digit recognition using spike-timing-dependent plasticity," *Frontiers Comput. Neurosci.*, vol. 9, p. 99, Aug. 2015, doi: [10.3389/fncom.2015.00099](https://doi.org/10.3389/fncom.2015.00099).